

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283334

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Sujan; Adith

INSULATION SYSTEM, AND ASSOCIATED ROOFING SYSTEM

Abstract

An insulation system includes a first insulation panel having a thickness of 0.4 inches to 6 inches, a second insulation panel including a first side and an opposite second side, a third insulation panel, a first material contacting the first side of the second insulation panel, the first material connecting the first insulation panel to the second insulation panel, and a second material contacting the second side of the second insulation panel, the second material connecting the second insulation panel to the third insulation panel, the insulation system positionable between a folded configuration in which the first insulation panel, the second insulation panel, and the third insulation panel are stacked, and an unfolded configuration in which the first insulation panel, the second insulation panel, and the third insulation panel are unstacked.

Inventors: Sujan; Adith (Mechanicsburg, PA)

Applicant: BMIC LLC (Dallas, TX)

Family ID: 1000008529196

Appl. No.: 19/073841

Filed: March 07, 2025

Related U.S. Application Data

us-provisional-application US 63562849 20240308

Publication Classification

Int. Cl.: E04D1/00 (20060101); E04D1/16 (20060101); E04D1/20 (20060101)

U.S. Cl.:

CPC E04D1/2942 (20190801); E04B1/80 (20130101); E04C2/205 (20130101); E04C2/405

Background/Summary

FIELD

[0001] The present invention relates to an insulation system and associated roofing system, and, more specifically, to an insulation system and associated roofing system including insulation panels.

BACKGROUND

[0002] A known roofing system includes multiple insulation panels installed on a roof deck.

SUMMARY

[0003] The Claims, rather than the Summary, define covered embodiments of the present invention. The Summary is a high-level overview of various aspects of the invention, and introduces concepts that are further described in the Detailed Description below. The Summary is not intended to identify key or essential features of the claimed subject matter, and also is not intended to be used in isolation to determine the scope of the claimed subject matter. Instead, the claimed subject matter should be understood by reference to appropriate portions of the Specification and drawings, as well as to each claim.

[0004] In some embodiments, the present invention provides an insulation system, comprising: a first insulation panel, wherein the first insulation panel has a thickness of 0.4 inches to 6 inches; a second insulation panel, wherein the second insulation panel includes a first side and an opposite second side; a third insulation panel; a first material contacting the first side of the second insulation panel, wherein the first material connects the first insulation panel to the second insulation panel; and a second material contacting the second side of the second insulation panel, wherein the second material connects the second insulation panel to the third insulation panel, wherein the insulation system is positionable between: a folded configuration in which the first insulation panel, the second insulation panel, and the third insulation panel are stacked, and an unfolded configuration in which the first insulation panel, the second insulation panel, and the third insulation panel are unstacked.

[0005] In some embodiments, a thickness of at least one of the second insulation panel or the third insulation panel is 0.5 inches to 4.6 inches.

[0006] In some embodiments, a width of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 4 feet, and a length of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 8 feet.

[0007] In some embodiments, at least one of the first material or the second material comprises double-sided tape.

[0008] In some embodiments, at least one of the first material or the second material comprises polyethylene.

[0009] In some embodiments, the first insulation panel comprises: a first layer, a second layer, and a third layer, wherein the second layer is between the first layer and the third layer, wherein the second layer comprises polyisocyanurate foam.

[0010] In some embodiments, the first insulation panel comprises: a first layer, a second layer, and a third layer, wherein the second layer is between the first layer and the third layer, wherein at least one of the first layer or the third layer comprises glass fiber-reinforced cellulosic felt.

[0011] In some embodiments, each of the first insulation panel, the second insulation panel, and the third insulation panel comprises a middle layer between two face layers. In some embodiments, at least one of the middle layers comprises polyisocyanurate foam.

[0012] In some embodiments, at least one of the face layers comprises felt.

[0013] In some embodiments, the present invention provides a roofing system, comprising: a roof deck; and an insulation system installed on the roof deck, wherein the insulation system comprises: a first insulation panel, wherein the first insulation panel has a thickness of 0.4 inches to 6 inches; a second insulation panel, wherein the second insulation panel includes a first side and an opposite second side; a third insulation panel; a first material contacting the first side of the second insulation panel, wherein the first material connects the first insulation panel to the second insulation panel; and a second material contacting the second side of the second insulation panel, wherein the second material connects the second insulation panel to the third insulation panel.

[0014] In some embodiments, the roof deck comprises at least one of metal, concrete, or wood.

[0015] In some embodiments, a thickness of at least one of the second insulation panel or the third insulation panel is 0.5 inches to 4.6 inches.

[0016] In some embodiments, a width of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 4 feet, and a length of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 8 feet.

[0017] In some embodiments, at least one of the first material or the second material comprises double-sided tape.

[0018] In some embodiments, the roofing system further comprises a membrane covering at least a portion of the insulation system, wherein the membrane comprises a single-ply membrane.

[0019] In some embodiments, the membrane comprises thermoplastic polyolefin (TPO).

[0020] In some embodiments, each of the first insulation panel, the second insulation panel, and the third insulation panel comprises a middle layer between two face layers. In some embodiments, at least one of the middle layers comprises polyisocyanurate foam.

[0021] In some embodiments, at least one of the face layers comprises felt.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] This section refers to the drawings that form a part of this disclosure, and which illustrate some of the embodiments of structure, materials, and/or methods of the present invention described herein.

[0023] FIG. 1 is a schematic cross-sectional view of an insulation system in an unfolded configuration installed on a roof deck, in accordance with some embodiments of the invention.

[0024] FIG. 2 is a schematic cross-sectional view on the insulation system of FIG. 1, in a folded configuration, in accordance with some embodiments of the invention.

DETAILED DESCRIPTION

[0025] In addition to the benefits and improvements that the Specification discloses, other objects and advantages that the Specification provides will become apparent from the following description taken in conjunction with the accompanying figures. Although the description discloses and describes detailed embodiments of the present disclosure, the disclosed embodiments are merely illustrative of the disclosure that may be embodied in various forms. In addition, each of the examples given regarding the various embodiments of the disclosure are intended to be illustrative, and not restrictive.

[0026] In some embodiments, the present invention provides an insulation system, including multiple insulation panels connected to one another, and configured to fold between a folded configuration in which the insulation panels are stacked on and/or in contact with one another, and an unfolded configuration in which the insulation panels are unstacked and/or out of contact with one another.

[0027] In some embodiments, one or more of the insulation panels include a first layer, a second layer, and a third layer, wherein the second layer is between the first layer and the third layer.

[0028] In some embodiments, one or more of the layers includes a side surface.

[0029] In some embodiments, the surface of one or more of the panels is substantially flat. In some embodiments, the surface of one or more of the panels is a taper surface.

[0030] In some embodiments, one or more of the insulation panels has a first edge, wherein the first edge is an edge of one of the layers.

[0031] In some embodiments, the one or more insulation panels includes a first insulation panel, which has a first layer, a second layer, and a third layer, wherein the second layer is between the first layer and the third layer. In some embodiments, the first insulation panel has a first edge, wherein the first edge is an edge of the first layer.

[0032] In some embodiments, the one or more insulation panels includes a second insulation panel. In some embodiments, the second insulation panel includes a fourth layer, a fifth layer, and a sixth layer, wherein the fifth layer is between the fourth layer and the sixth layer. In some embodiments, the second insulation panel has a second edge and a third edge, wherein the second edge is an edge of the fourth layer, and wherein the third edge is an edge of the sixth layer.

[0033] In some embodiments, the one or more insulation panels includes a third insulation panel. In some embodiments, the third insulation panel includes a seventh layer, an eighth layer, and a ninth layer, wherein the eighth layer is between the seventh layer and the ninth layer. In some embodiments, the third insulation panel has a fourth edge, wherein the fourth edge is an edge of the ninth layer.

[0034] In some embodiments, a middle layer of at least one of the insulation panels includes polyisocyanurate. In some embodiments, the middle layer includes polyisocyanurate foam. In some embodiments, at least one of the second layer, the fifth layer, or the eighth layer includes polyisocyanurate foam. In some embodiments, two or more of the second layer, the fifth layer, or the eighth layer includes polyisocyanurate foam. In some embodiments, the second layer, the fifth layer, and the eighth layer includes polyisocyanurate foam.

[0035] In some embodiments, one or more layers of the panels includes a coverboard. In some embodiments, the coverboard is between the foof insulation and the membrane. In some embodiments, the coverboard is composed of fiberglass-faced gypsum, high-density polyiso, perlite, cementitious wood fiber, and/or coated glass mat boards.

[0036] In some embodiments, a top or uppermost layer, and/or a bottom or lowermost layer, of at least one of the insulation panels includes glass fiber-reinforced cellulosic felt. In some embodiments, at least one of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes glass fiber-reinforced cellulosic felt. In some embodiments, two or more of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes glass fiber-reinforced cellulosic felt.

[0037] In some embodiments, the top or uppermost layer, and/or the bottom or lowermost layer, of at least one of the insulation panels includes a coated glass fiber mat. In some embodiments, at least one of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes a coated glass fiber mat. In some embodiments, two or more of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes a coated glass fiber mat.

[0038] In some embodiments, the top or uppermost layer, and/or the bottom or lowermost layer, of one or more of the insulation panels includes aluminum foil. In some embodiments, at least one of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes aluminum foil. In some embodiments, two or more of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes aluminum foil.

[0039] In some embodiments, the top or uppermost layer, and/or the bottom or lowermost layer, of one or more of the insulation panels includes two or more of glass fiber-reinforced cellulosic felt, coated glass fiber mat, and/or aluminum foil. In some embodiments, at least one of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes

aluminum foil. In some embodiments, two or more of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer includes aluminum foil.

[0040] In some embodiments, the top or uppermost layer, and/or the bottom or lowermost layer, of one or more of the insulation panels includes two or more sublayers. In some embodiments, the top or uppermost layer, and/or the bottom or lowermost layer, of the insulation panels includes three or more sublayers.

[0041] In some embodiments, the insulation system includes a first material. In some embodiments, the first material contacts, connects to, and/or at least partially overlaps the first edge and the second edge. In some embodiments, the contact is direct, with no intervening layer therebetween. In some embodiments, the contact is indirect, with one or more intervening layers therebetween.

[0042] In some embodiments, the insulation system includes a second material. In some embodiments, the second material contacts, connects to, and/or at least partially overlaps the third edge and the fourth edge. In some embodiments, the contact is direct, with no intervening layer therebetween. In some embodiments, the contact is indirect, with one or more intervening layers therebetween.

[0043] In some embodiments, there is sufficient spacings between the first edge and the second edge, and between the second edge and the third edge, which are configured to permit the insulation system to fold between the folded configuration in which the first insulation panel, the second insulation panel, and/or the third insulation panel are stacked on and/or in contact with one another, and the unfolded configuration in which the first insulation panel, the second insulation panel, and/or the third insulation panel are unstacked and/or out of contact with one another.

[0044] In some embodiments, the insulation panels are connected to neighboring insulation panels, thereby creating a fanfold configuration.

[0045] In some embodiments, the insulation panels of the insulation system are a bundle when in the folded configuration.

[0046] In some embodiments, at least one of the first material or the second material includes tape. In some embodiments, at least one of the first material or the second material includes double-sided tape. In some embodiments, the double-sided tape includes a removable liner. In some embodiments, the removable liner is removed to expose an adhesive. In some embodiments, the adhesive adheres the insulation system to the roof deck.

[0047] In some embodiments, at least one of the first material or the second material includes polyethylene.

[0048] In some embodiments, at least one of the first material or the second material is a vapor retarder membrane.

[0049] In some embodiments, the present invention provides a roofing system, including a roof deck, and an insulation system installed on the roof deck.

[0050] In some embodiments, the roof deck is a flat roof deck.

[0051] In some embodiments, the roof deck is a low slope roof deck. As used herein, a “low slope” roof or roofing substrate is a roof or roofing substrate having a pitch of Y/X, where Y and X are in a ratio of at most 2:12, where Y corresponds to the “rise” of the roof or roof deck, and where X corresponds to the “run” of the roof or roof deck.

[0052] In some embodiments, the roof deck is a roof deck of a commercial building.

[0053] In some embodiments, the roof deck includes a first portion, a second portion, and a third portion. In some embodiments, the first insulation panel is installed on the first portion of the roof deck. In some embodiments, the bottom layer of the first panel is juxtaposed with the first portion of the roof deck. In some embodiments, the first layer and/or face of the first panel is juxtaposed with the first portion of the roof deck.

[0054] In some embodiments, the roofing system includes the second insulation panel installed on the second portion of the roof deck. In some embodiments, the bottom layer and/or face of the

second panel is juxtaposed with the second portion of the roof deck.

[0055] In some embodiments, the fourth layer of the second panel is juxtaposed with the second portion of the roof deck.

[0056] In some embodiments, the roofing system includes the third insulation panel installed on the third portion of the roof deck. In some embodiments, the bottom layer and/or face of the third panel is juxtaposed with the third portion of the roof deck. In some embodiments, the seventh layer of the third panel is juxtaposed with the third portion of the roof deck.

[0057] In some embodiments, the roofing system includes the first material. In some embodiments, the first material contacts, connects to, and/or at least partially overlaps the first edge and the second edge. In some embodiments, the first material is juxtaposed with the roof deck.

[0058] In some embodiments, the roofing system includes the second material. In some embodiments, the second material contacts, connects to, and/or at least partially overlaps the third edge and the fourth edge.

[0059] In some embodiments, the roof deck includes at least one of metal, such as but not limited to steel, concrete, or wood.

[0060] In some embodiments, the roofing system includes an existing roof. In some embodiments, the existing roof includes a membrane.

[0061] In some embodiments, the roofing system further includes a membrane covering at least a portion of the first insulation panel, the second insulation panel, and the third insulation panel. In some embodiments, the membrane covers at least a portion of the second material.

[0062] In some embodiments, the membrane includes a single-ply membrane.

[0063] In some embodiments, the membrane includes thermoplastic polyolefin (TPO). In some embodiments, the membrane includes polyvinyl chloride (PVC).

[0064] In some embodiments, the roofing system omits a separate vapor retarder membrane, other than the one or more materials connecting one or more of the insulation panels to one another.

[0065] In some embodiments, one or more of the insulation panels has a thickness of 0.3 inches. In some embodiments, the thickness is 0.4 inches. In some embodiments, the thickness is 0.5 inches. In some embodiments, the thickness is 0.6 inches. In some embodiments, the thickness is 0.7 inches. In some embodiments, the thickness is 0.8 inches. In some embodiments, the thickness is 0.9 inches. In some embodiments, the thickness is 1 inch. In some embodiments, the thickness is 1.1 inch. In some embodiments, the thickness is 1.2 inches. In some embodiments, the thickness is 1.3 inches. In some embodiments, the thickness is 1.4 inches. In some embodiments, the thickness is 1.5 inches. In some embodiments, the thickness is 2 inches. In some embodiments, the thickness is 2.5 inches. In some embodiments, the thickness is 3 inches. In some embodiments, the thickness is 3.5 inches. In some embodiments, the thickness is 4 inches. In some embodiments, the thickness is 4.5 inches. In some embodiments, the thickness is 4.6 inches. In some embodiments, the thickness is 4.7 inches. In some embodiments, the thickness is 4.8 inches. In some embodiments, the thickness is 4.9 inches. In some embodiments, the thickness is 5 inches. In some embodiments, the thickness is 5.5 inches. In some embodiments, the thickness is 6 inches. In some embodiments, the thickness is greater than 6 inches.

[0066] In some embodiments, the thickness is 0.3 to 4.7 inches. In some embodiments, the thickness is 0.4 to 4.7 inches. In some embodiments, the thickness is 0.5 to 4.7 inches. In some embodiments, the thickness is 0.6 to 4.7 inches. In some embodiments, the thickness is 0.7 to 4.7 inches. In some embodiments, the thickness is 0.8 to 4.7 inches. In some embodiments, the thickness is 0.9 to 4.7 inches. In some embodiments, the thickness is 1 to 4.7 inches. In some embodiments, the thickness is 1.5 to 4.7 inches.

[0067] In some embodiments, the thickness is 2 to 4.7 inches. In some embodiments, the thickness is 2.5 to 4.7 inches. In some embodiments, the thickness is 3 to 4.7 inches. In some embodiments, the thickness is 3.5 to 4.7 inches. In some embodiments, the thickness is 4 to 4.7 inches. In some embodiments, the thickness is 4.5 to 4.7 inches. In some embodiments, the thickness is 4.6 to 4.7

[0076] In some embodiments, the thickness is 0.3 to 2 inches. In some embodiments, the thickness is 0.4 to 2 inches. In some embodiments, the thickness is 0.5 to 2 inches. In some embodiments, the thickness is 0.6 to 2 inches. In some embodiments, the thickness is 0.7 to 2 inches. In some embodiments, the thickness is 0.8 to 2 inches. In some embodiments, the thickness is 0.9 to 2 inches. In some embodiments, the thickness is 1 to 2 inches. In some embodiments, the thickness is 1.5 to 2 inches.

[0077] In some embodiments, the thickness is 0.3 to 1.5 inches. In some embodiments, the thickness is 0.4 to 1.5 inches. In some embodiments, the thickness is 0.5 to 1.5 inches. In some embodiments, the thickness is 0.6 to 1.5 inches. In some embodiments, the thickness is 0.7 to 1.5 inches. In some embodiments, the thickness is 0.8 to 1.5 inches. In some embodiments, the thickness is 0.9 to 1.5 inches. In some embodiments, the thickness is 1 to 1.5 inches.

[0078] In some embodiments, at least two of the insulation panels are the same thickness. In some embodiments, two of the insulation panels are the same thickness. In some embodiments, three of the insulation panels are the same thickness. In some embodiments, four of the insulation panels are the same thickness. In some embodiments, five of the insulation panels are the same thickness. In some embodiments, six of the insulation panels are the same thickness. In some embodiments, seven of the insulation panels are the same thickness. In some embodiments, eight of the insulation panels are the same thickness. In some embodiments, nine of the insulation panels are the same thickness. In some embodiments, ten of the insulation panels are the same thickness. In some embodiments, more than ten of the insulation panels are the same thickness. In some embodiments, all of the insulation panels of the insulation system are the same thickness.

[0079] In some embodiments, at least two of the insulation panels are different thicknesses. In some embodiments, two of the insulation panels are different thicknesses. In some embodiments, three of the insulation panels are different thicknesses. In some embodiments, four of the insulation panels are different thicknesses. In some embodiments, five of the insulation panels are different thicknesses. In some embodiments, six of the insulation panels are different thicknesses. In some embodiments, seven of the insulation panels are different thicknesses. In some embodiments, eight of the insulation panels are different thicknesses. In some embodiments, nine of the insulation panels are different thicknesses. In some embodiments, ten of the insulation panels are different thicknesses. In some embodiments, more than ten of the insulation panels are different thicknesses. In some embodiments, all of the insulation panels of the insulation system are different thicknesses.

[0080] In some embodiments, one or more the insulation panels are tapered—e.g., does not have a constant thickness.

[0081] In some embodiments, at least one of the insulation panels has a width of 2 feet. In some embodiments, the width is 3 feet. In some embodiments, the width is 4 feet. In some embodiments, the width is 5 feet. In some embodiments, the width is 6 feet.

[0082] In some embodiments, the width is 2 to 6 feet. In some embodiments, the width is 3 to 6 feet. In some embodiments, the width is 4 to 6 feet. In some embodiments, the width is 5 to 6 feet.

[0083] In some embodiments, the width is 2 to 5 feet. In some embodiments, the width is 3 to 5 feet. In some embodiments, the width is 4 to 5 feet.

[0084] In some embodiments, the width is 2 to 4 feet. In some embodiments, the width is 3 to 4 feet. In some embodiments, the width is 2 to 3 feet.

[0085] In some embodiments, at least two of the panels are the same width. In some embodiments, two of the panels are the same width. In some embodiments, three of the panels are the same width. In some embodiments, four of the panels are the same width. In some embodiments, five of the panels are the same width. In some embodiments, six of the panels are the same width. In some embodiments, seven of the panels are the same width. In some embodiments, eight of the panels are the same width. In some embodiments, nine of the panels are the same width. In some embodiments, ten of the panels are the same width. In some embodiments, more than ten of the panels are the same width. In some embodiments, all of the panels of the insulation system are the

same width.

[0086] In some embodiments, at least two of the panels are different widths. In some embodiments, two of the panels are different widths. In some embodiments, three of the panels are different widths. In some embodiments, four of the panels are different widths. In some embodiments, five of the panels are different widths. In some embodiments, six of the panels are different widths. In some embodiments, seven of the panels are different widths. In some embodiments, eight of the panels are different widths. In some embodiments, nine of the panels are different widths. In some embodiments, ten of the panels are different widths. In some embodiments, more than ten of the panels are different widths. In some embodiments, all of the panels of the insulation system are different widths.

[0087] In some embodiments, at least one of the insulation panels has a length of 2 feet. In some embodiments, at least one of the insulation panels has a length of 3 feet. In some embodiments, at least one of the insulation panels has a length of 4 feet. In some embodiments, at least one of the insulation panels has a length of 5 feet. In some embodiments, at least one of the insulation panels has a length of 6 feet. In some embodiments, the length is 7 feet. In some embodiments, the length is 8 feet. In some embodiments, the length is 9 feet. In some embodiments, the length is 10 feet.

[0088] In some embodiments, the length is 3 to 9 feet. In some embodiments, the length is 4 to 9 feet. In some embodiments, the length is 5 to 9 feet. In some embodiments, the length is 6 to 9 feet. In some embodiments, the length is 7 to 9 feet. In some embodiments, the length is 8 to 9 feet. In some embodiments, the length is 3 to 8 feet. In some embodiments, the length is 4 to 8 feet. In some embodiments, the length is 5 to 8 feet. In some embodiments, the length is 6 to 8 feet. In some embodiments, the length is 7 to 8 feet.

[0089] In some embodiments, the length is 3 to 7 feet. In some embodiments, the length is 4 to 7 feet. In some embodiments, the length is 5 to 7 feet. In some embodiments, the length is 6 to 7 feet. In some embodiments, the length is 3 to 6 feet. In some embodiments, the length is 4 to 6 feet. In some embodiments, the length is 5 to 6 feet. In some embodiments, the length is 3 to 5 feet. In some embodiments, the length is 4 to 5 feet. In some embodiments, the length is 3 to 4 feet.

[0090] In some embodiments, one or more of the materials are along the width edges of one or more of the insulation panels. In some embodiments, one or more of the materials are along the length edges of one or more of the insulation panels.

[0091] In some embodiments, the insulation system includes more than three insulation panels. In some embodiments, the insulation system includes four insulation panels. In some embodiments, the insulation system includes five insulation panels. In some embodiments, the insulation system includes six insulation panels. In some embodiments, the insulation system includes seven insulation panels. In some embodiments, the insulation system includes eight insulation panels. In some embodiments, the insulation system includes nine insulation panels. In some embodiments, the insulation system includes ten insulation panels. In some embodiments, the insulation system includes more than ten insulation panels.

[0092] In some embodiments, at least two of the panels are the same length. In some embodiments, two of the panels are the same length. In some embodiments, three of the panels are the same length. In some embodiments, four of the panels are the same length. In some embodiments, five of the panels are the same length. In some embodiments, six of the panels are the same length. In some embodiments, seven of the panels are the same length. In some embodiments, eight of the panels are the same length. In some embodiments, nine of the panels are the same length. In some embodiments, ten of the panels are the same length. In some embodiments, more than ten of the panels are the same length. In some embodiments, all of the panels of the insulation system are the same length.

[0093] In some embodiments, at least two of the panels are different lengths. In some embodiments, two of the panels are different lengths. In some embodiments, three of the panels are different lengths. In some embodiments, four of the panels are different lengths. In some

embodiments, five of the panels are different lengths. In some embodiments, six of the panels are different lengths. In some embodiments, seven of the panels are different lengths. In some embodiments, eight of the panels are different lengths. In some embodiments, nine of the panels are different lengths. In some embodiments, ten of the panels are different lengths. In some embodiments, more than ten of the panels are different lengths. In some embodiments, all of the panels of the insulation system are different lengths.

[0094] With reference to the drawings, FIG. 1 is a schematic cross-sectional view of a roofing system including an insulation system **1000** in an unfolded configuration installed on a roof deck **800**, and FIG. 2 is a schematic cross-sectional view on the insulation system in a folded configuration, in accordance with some embodiments of the invention. In some embodiments, as FIGS. 1 and 2 illustrate, the insulation system includes a first insulation panel **100**, a second insulation panel **200**, and a third insulation panel **300**. In some embodiments, the first insulation panel **100** includes a first layer **101**, a second layer **102**, and a third layer **103**, and the second layer **102** is between the first layer **101** and the third layer **103**, as FIGS. 1 and 2 show. In some embodiments, the first insulation panel **100** has a first edge **110**. In some embodiments, the first edge **110** is an edge of the first layer **101**. In some embodiments, first insulation panel **100** includes a top side **111**, and a bottom side **112** opposite the top side **111**.

[0095] In some embodiments, the second insulation panel **200** includes a fourth layer **204**, a fifth layer **205**, and a sixth layer **206**. The fifth layer **205** is between the fourth layer **204** and the sixth layer **206**. In some embodiments, the second insulation panel **200** has a second edge **220** and a third edge **230**. In some embodiments, the second edge **220** is an edge of the fourth layer **204**. In some embodiments, the third edge **230** is an edge of the sixth layer **206**. In some embodiments, second insulation panel **200** includes a top side **211**, and a bottom side **212** opposite the top side **211**.

[0096] As FIGS. 1 and 2 show, in some embodiments, the third insulation panel **300** includes a seventh layer **307**, an eighth layer **308**, and a ninth layer **309**, and the eighth layer **308** is between the seventh layer **307** and the ninth layer **309**. In some embodiments, the third insulation panel **300** has a fourth edge **340**, and the fourth edge **340** is an edge of the ninth layer **309**. In some embodiments, third insulation panel **300** includes a top side **311**, and a bottom side **312** opposite the top side **311**.

[0097] In some embodiments, as FIGS. 1 and 2 illustrate, the insulation system **1000** includes a first material **400**. In some embodiments, the first material **400** contacts, connects to, and/or at least partially overlaps the first edge **110** of the first insulation panel **100** and the second edge **220** of the second insulation panel **200**. In some embodiments, the first material **400** contacts, connects to, and/or at least partially overlaps the bottom side **112** of the first insulation panel **100** and the bottom side **212** of the second insulation panel **200**.

[0098] As FIGS. 1 and 2 show, in some embodiments, the insulation system **1000** includes a second material **500**. In some embodiments, the second material **500** contacts, connects to, and/or at least partially overlaps the third edge **230** of the second insulation panel **200** and the fourth edge **340** of the third insulation panel **300**. In some embodiments, the second material **500** contacts, connects to, and/or at least partially overlaps the top side **211** of the second insulation panel **200** and the top side **311** of the third insulation panel **300**. As the figures show, in some embodiments, the insulation system **1000** is positionable between a folded configuration in which the first insulation panel **100**, the second insulation panel **200**, and the third insulation panel **300** are stacked on one another, as FIG. 2 illustrates, and an unfolded configuration in which the first insulation panel **100**, the second insulation panel **200**, and the third insulation panel **300** are unstacked and out of contact with one another, as FIG. 1 illustrates.

[0099] In some embodiments, as FIG. 1 shows, the roofing system includes a membrane **900**. In some embodiments, as the figure shows, the membrane **900** is installed over the insulation system **1000**.

[0100] It is to be understood that, in some embodiments, the insulation system **1000** includes more

than three insulation panels, and more than two materials. It is to be further understood that, in some embodiments, the materials are alternately attached to upper and lower faces, sides, and/or layers of the insulation panels—e.g., in some embodiments, a material connects to the tops of a first insulation panel and a second insulation panel, another material connects to the bottoms of the second insulation panel and a third insulation panel, another material connects to the tops of the third insulation panel and a fourth insulation panel, etc., with tops and bottoms being relative terms describing the orientation of the insulation system, for example, when not installed, such that the insulation system **1000** may be reoriented with the former bottom being the top, and vice versa.

[0101] Variations, modifications, and alterations to embodiments of the present disclosure described above will make themselves apparent to those skilled in the art. All such variations, modifications, alterations and the like are intended to fall within the spirit and scope of the present disclosure, limited solely by the appended claims.

[0102] While several embodiments of the present disclosure have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. For example, all dimensions discussed herein are provided as examples only, and are intended to be illustrative and not restrictive.

[0103] Any feature or element that is positively identified in this description may also be specifically excluded as a feature or element of an embodiment of the present as defined in the claims.

[0104] The disclosure described herein may be practiced in the absence of any element or elements, limitation or limitations, which is not specifically disclosed herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the disclosure.

Claims

1. A roofing system, comprising: a roof deck; an insulation system installed on the roof deck, the insulation system comprising: a first insulation panel having a thickness ranging from 0.4 inches to 6 inches; a second insulation panel having a first side and an opposite second side; a third insulation panel; a first material contacting the first side of the second insulation panel, wherein the first material connects the first insulation panel to the second insulation panel; a second material contacting the second side of the second insulation panel, wherein the second material connects the second insulation panel to the third insulation panel; wherein the insulation system is positionable between: a folded configuration in which the first insulation panel, the second insulation panel, and the third insulation panel are stacked, and an unfolded configuration in which the first insulation panel, the second insulation panel, and the third insulation panel are unstacked.
2. The roofing system of claim 1, wherein the roof deck is made of metal, concrete, or wood.
3. The roofing system of claim 1, wherein the first insulation panel includes a middle layer of polyisocyanurate foam.
4. The roofing system of claim 3, wherein the first insulation panel has face layers made of glass fiber-reinforced cellulosic felt.
5. The roofing system of claim 1, wherein the second insulation panel includes a middle layer of polyisocyanurate foam.
6. The roofing system of claim 5, wherein the second insulation panel has face layers made of a coated glass fiber mat.
7. The roofing system of claim 1, wherein the first material comprises double-sided tape with a removable liner to expose an adhesive.
8. The roofing system of claim 1, wherein the second material comprises a vapor retarder

membrane.

- 9.** The insulation system of claim 1, wherein a length of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 4 feet, and wherein a width of at least one of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 8 feet.
 - 10.** The insulation system of claim 1, wherein at least one of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer comprises glass fiber-reinforced cellulosic felt.
 - 11.** The insulation system of claim 1, wherein at least one of the first membrane or the second membrane comprises polyethylene.
 - 12.** The insulation system of claim 7, wherein at least one of the first layer, the third layer, the fourth layer, the sixth layer, the seventh layer, or the ninth layer comprises glass fiber-reinforced cellulosic felt.
 - 13.** A roofing system, comprising: a roof deck, wherein the roof deck includes a first portion, a second portion, and a third portion; a first insulation panel installed above the first portion of the roof deck, wherein the first insulation panel comprises: a first layer, a second layer, and a third layer, wherein the first layer is juxtaposed with the first portion, wherein the second layer is between the first layer and the third layer, wherein the first insulation panel has a first edge, wherein the first edge is an edge of the first layer, wherein the first insulation panel has a thickness of 0.4 inches to 6 inches; a second insulation panel, wherein the second insulation panel comprises: a fourth layer, a fifth layer, and a sixth layer, wherein the fourth layer is juxtaposed with the second portion, wherein the fifth layer is between the fourth layer and the sixth layer, wherein the second insulation panel has a second edge and a third edge, wherein the second edge is an edge of the fourth layer, wherein the third edge is an edge of the sixth layer, wherein the second insulation panel has a thickness of 0.4 inches to 6 inches; a third insulation panel, wherein the third insulation panel comprises: a seventh layer, an eighth layer, and a ninth layer, wherein the seventh layer is juxtaposed with the third portion, wherein the eighth layer is between the seventh layer and the ninth layer, wherein the third insulation panel has a fourth edge, wherein the fourth edge is an edge of the ninth layer, wherein the third insulation panel has a thickness of 0.4 inches to 6 inches; a first membrane, wherein the first membrane connects to and overlaps the first edge and the second edge, wherein the first membrane is juxtaposed with the roof deck; and a second membrane, wherein the second membrane connects to and overlaps the third edge and the fourth edge.
 - 14.** The roofing system of claim 11, wherein the roof deck comprises at least one of metal, concrete, or wood.
 - 15.** The roofing system of claim 11, wherein a thickness of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 0.5 inches to 4.6 inches.
 - 16.** The roofing system of claim 11, wherein a length of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 4 feet, and wherein a width of at least one of the first insulation panel, the second insulation panel, or the third insulation panel is 8 feet.
 - 17.** The roofing system of claim 11, wherein at least one of the first membrane or the second membrane comprises double-sided tape.
 - 18.** The roofing system of claim 11, further comprising: a third membrane covering at least a portion of the first insulation panel, the second insulation panel, and the third insulation panel, wherein the third membrane comprises a single-ply membrane.
 - 19.** The roofing system of claim 16, wherein the third membrane comprises thermoplastic polyolefin (TPO).
 - 20.** The roofing system of claim 11, wherein at least one of the second layer, the fifth layer, or the seventh layer comprises polyisocyanurate.
-